

Explainable AI for Predicting Cybersecurity Vulnerabilities in Software Systems: Toward Trustworthy and Transparent Cybersecurity

I. Abstract

Software systems have been facing a very fast growing number of security vulnerabilities. Machine learning has become widely used to predict and rank these vulnerabilities. These ML models are often accurate but act as black boxes, which makes security analysts unsure whether to trust their results. Explainable Artificial Intelligence or XAI has been offered as a solution. But evidence has shown that explanations raise trust and can even be attacked. This proposal examines how explainable artificial intelligence can make the prediction of software vulnerabilities in a more transparent and more trustworthy way. By going through a conceptual review of eleven recent studies, it compares how explanation methods are applied, how they support transparency and trust, and how robust they are against distortion. The study identifies a gap between prediction research and trust research, and proposes a conceptual framework that links prediction, explanation quality, transparency, and trust and by this it is offering a base for future empirical testing on real vulnerability data.

Index Terms. Explainable AI, vulnerability prediction, software security, SHAP, LIME, transparency, trust, conceptual review.

II. Introduction

Artificial Intelligence is something that Cybersecurity is very much relying on nowadays, because the number of software vulnerabilities and attacks has grown beyond what human teams can handle alone [8], [10]. These can be predicted so quickly by machine learning models, but decisions are often much harder to understand. This lack of clarity makes security analysts unsure whether they can trust the results [9]. Explainable artificial intelligence(XAI) has been offered as the answer, since it tries to show the reasons behind a model decision so that people can understand and interpret it [1], [5].

This proposal studies how XAI can make the prediction of software vulnerabilities more transparent and more trustworthy. It does this as a conceptual review of eleven recent studies, drawing them together to see how the pieces can be connected. The rest of the proposal is organised as follows: The background gives the context. The literature review examines the existing studies to find the research gap. The problem statement defines what is missing and why it matters. The aim, the objectives, and the research questions set the direction to a conceptual framework and a methodology. Then it is explained how the study will be carried out, followed by the expected contribution, the ethical considerations, a timeline, and a conclusion.

III. Background

Software systems have come to support almost every part of modern life, from banking and healthcare to power grids, transport, and personal devices. As these systems have grown larger and intensely connected, the number of security vulnerabilities found in them has been tremendously rising every year. A vulnerability is a weakness that an attacker could use to gain access or cause harm, and each one must be recorded, studied, and given a severity rating before it can be fixed [1]. Public records such as the National Vulnerability Database have shown a steady and sharp rise in reported vulnerabilities over the past two decades. Thus the effort needed to assess them has grown just as fast [1].

The volume has become so large that security teams can no longer review every weakness manually by hand. This has pushed the field toward automated help [8], [10]. To cope with the large scale, researchers have started increasingly using machine learning models to predict which vulnerabilities exist, how serious they are, and which threats are most urgent [1], [6], [10]. These models can reach higher accuracy and can process far more data than a human analyst.

One review has reported that AI can raise the accuracy of threat detection by a large margin and can cut response times significantly, which shows why the field has moved in this direction [10]. However, many of the strongest models work as black boxes, meaning they give an output result without even showing the reason behind it. In the security analysis field this has been a serious limit, reason is an analyst who must patch a system, raise an alarm, or warn a client needs to understand why a prediction was made before even acting on it [9]. A score with no reason is hard to defend to a manager, an auditor, or a customer, and it is also hard to check for errors. Thus the need for understanding in depth is the reason which has given rise to the field of explainable artificial intelligence or XAI.

XAI offers methods that show which features pushed a model towards its decision or output. Two of the most common are SHAP and LIME, which highlight the inputs that mattered most for a single prediction, so the result becomes easier to read, check and question [1], [5], [6]. The aim is to make ML model decisions more transparent for the people who rely on them so cybersecurity applies these techniques to tasks or jobs such as malware analysis, intrusion detection, and putting scores on software vulnerabilities [5], [6], [9].

Few recent analyses have warned that explanations are not automatically reliable. A large study that combined the results of many earlier works, has found that explanations raise user trust only to a moderate degree [7]. Other research has shown that the explanations themselves can be attacked or quietly deformed by an adversary and because of that a harmful decision can be made to look reasonable [2], [9]. All these anomalies have created a tension at the heart of the field. Explainable methods promise clear and trustworthy predictions, but the clarity they offer may be weaker, or less safe, than it appears. This proposal enters that space, focusing on the prediction of software vulnerabilities and on whether explainable methods can make those predictions both transparent and trustworthy.

IV. Literature Review

Four ideas sit at the centre of this review. The first is vulnerability prediction, meaning the use of a ML model to decide whether a software weakness exists or how severe it is, so that teams can fix the most dangerous ones first [1]. The second is explainable artificial intelligence or XAI, a set of procedures that show why a model made a decision, with SHAP, LIME, and integrated gradients [1], [2], [6]. The third is transparency, which is the degree to which a person can see the motive or cause behind a prediction. The fourth is trust, which is the confidence a security analyst places in a prediction and in the explanation that comes with it [7]. A further idea, resilience or robustness, runs through the review and it describes how well an explanation holds up when someone tries to twist it [2], [9].

A first group of studies has applied explainable methods to real security tasks. The work closest to this proposal is by Manai and colleagues, who have built an interpretable model that predicts the severity metrics of software vulnerabilities from their text descriptions [1]. Rather than stopping at a simple importance picture, they have used SHAP across several classes at once, turned the values into cumulative word importance, and produced clear rules that a vulnerability analyst can act on. They have also released their data and code, and have reported that the approach can help analysts score vulnerabilities faster and with more confidence [1].

Other studies have applied the same explainable ideas to detecting attacks rather than predicting vulnerabilities. Alabri and colleagues have combined a generative model with an attention based network to detect low rate denial of service attacks in cloud systems, reaching high accuracy and using SHAP to name the traffic features that mattered most [3]. Hamim and colleagues have built a hybrid intrusion detection system that first filters normal traffic and then classifies threats with an ensemble of models, and they have added SHAP for an overall view and LIME for single alerts [6]. Manthena and colleagues have surveyed how explainable methods are used across malware analysis and have mapped the open challenges [5], while Li and colleagues have proposed a framework that joins robust learning, privacy protection, and attention based explanation [11]. Together these works show that explainable methods have become widely used and can reach strong results.

A second group has looked more critically at the explanations themselves. Zahra and Lewis have combined the results of ninety studies and have found that explainability is linked to trust only by a moderate amount, which means that adding an explanation does not guarantee that users will trust a system [7]. Mia and Pritom have shown that explanations can be attacked through several methods, so that a model may appear to give an honest reason while the reason has been quietly changed to hide a harmful decision [2]. Alshudukhi and colleagues have reviewed this double role of explainable methods, describing them as both a tool for transparency and a fresh target for attackers, and have listed defences such as robust training and privacy protection that are still developing [9].

A third group has given wider context. Vikalp Thapliyal and Pranita Thapliyal have reviewed machine learning across many cybersecurity tasks, including malware, intrusion, and botnet

detection [8]. Bhandari has examined the benefits, the risks, and the governance of artificial intelligence in security, and has reported large gains in accuracy and response speed alongside serious concerns about misuse and bias [10]. Carter and colleagues have discussed explainable artificial intelligence in corporate risk and security analytics, arguing that transparency supports trust, compliance, and better decisions inside companies [4].

When the three groups are compared, a clear pattern appears. The application studies treat an explanation as a benefit that improves transparency and helps the analyst act. The critical studies treat an explanation as something fragile that may mislead, either because it raises trust only a little or because it can be attacked. The wider reviews accept both the promise and the risk, but they discuss them in general terms rather than for any single task. The two main views, the hopeful one and the cautious one, have rarely met inside the same study.

Several limitations stand out across these works. Most of the application studies have focused on detecting attacks such as malware and breach/infiltration, while the prediction of software vulnerabilities has received far less attention. With the work of Manai being a rare exception [1], [3], [5], [6]. The studies that question trust and robustness have been kept separate from the studies that build prediction models, so the strengths of one are seldom tested against the warnings of the other [2], [7], [9]. Few studies have measured whether an explanation is faithful or whether it survives a deliberate threat or cyber attack, and fewer still have done so for vulnerability prediction in particular.

Table 1 summarises the reviewed papers, showing for each one its security task, its explanation method, its main finding, and its main limitation.

Reference	Security task	Explanation method	Main finding	Main limitation
[1] Manai et al.	Vulnerability severity scoring	SHAP with interpretable model and rules	Accurate, actionable severity scoring	Explanations not tested for robustness
[2] Mia and Pritom	Attacks on explanations (phishing, malware, intrusion, fraud)	SHAP, LIME, integrated gradients (attacked)	Explanations can be fooled or hidden	Not applied to vulnerability prediction
[3] Alabri et al.	Low rate DDoS detection in cloud	SHAP with generative and attention model	High accuracy, key flow features identified	Detection task, not vulnerability prediction
[4] Carter et al.	Corporate risk and security analytics	Explainable AI (conceptual)	Transparency supports trust and compliance	Conceptual, no empirical test

[5] Manthena et al.	Malware analysis (survey)	Various XAI methods	Maps XAI methods and open challenges	Survey on malware, not prediction
[6] Hamim et al.	Intrusion detection	SHAP and LIME with hybrid ensemble	High accuracy with clear explanations	Detection task, not vulnerability prediction
[7] Zahra Atf and Lewis	Trust and explainability (meta analysis)	Meta analysis of ninety studies	Explainability raises trust only moderately	Not specific to security or prediction
[8] Vikalp Thapliyal and Pranita Thapliyal	Machine learning for cybersecurity (review)	General review, limited XAI	Overview of ML for threat detection	Little focus on explainability
[9] Alshudukhi et al.	XAI in cybersecurity (review)	Review of XAI methods and attacks	XAI is both a transparency tool and an attack surface	General, not vulnerability prediction
[10] Bhandari	AI and cybersecurity governance (review)	Review of fifty studies	Gains in accuracy and speed, with risks	Governance focus, not explainable prediction
[11] Li et al.	Robust and private threat detection	FALCA with attention explanation	Resilient, private, interpretable detection	Detection task, not vulnerability prediction

Table 1. Summary of the eleven reviewed papers.

Taken together, these limitations show that the field has not yet joined two things that clearly belong together, namely the prediction of software vulnerabilities and the question of whether the explanations behind those predictions can be trusted. The next section sets out this gap in detail.

V. Research Gap

Literature review has revealed a very clear gap. Most existing work has applied explainable methods to detect cyber attacks while the specific task of predicting software vulnerabilities has rarely been studied through an explainability lens [1], [3], [5], [6]. At the same time, the studies

that question whether explanations can be trusted, and whether they survive deliberate attack, have formed a separate line of research from the studies that build the prediction models [2], [7], [9]. Because these two lines have been kept apart, the field which has strong prediction models, their explanations have not been tested for trust and robustness, No reviewed study has brought the two together for the prediction of software vulnerabilities.

This study addresses that missing connection. By reviewing the eleven selected papers as one body of research, it draws the prediction studies and the trust studies into a single view. So that the promise of transparent prediction can be weighed against the doubts about explanation quality and robustness. From this combined reading, this study reviews existing works and proposes a conceptual framework that links four ideas, namely vulnerability prediction, explanation quality, transparency, and trust, and thus shows how they influence one another for cybersecurity. The framework is built so that a later empirical study could test it on a real vulnerability dataset, such as the records held in the National Vulnerability Database that Manai and colleagues have used [1]. Figure 1 summarises this gap and how the study addresses it.

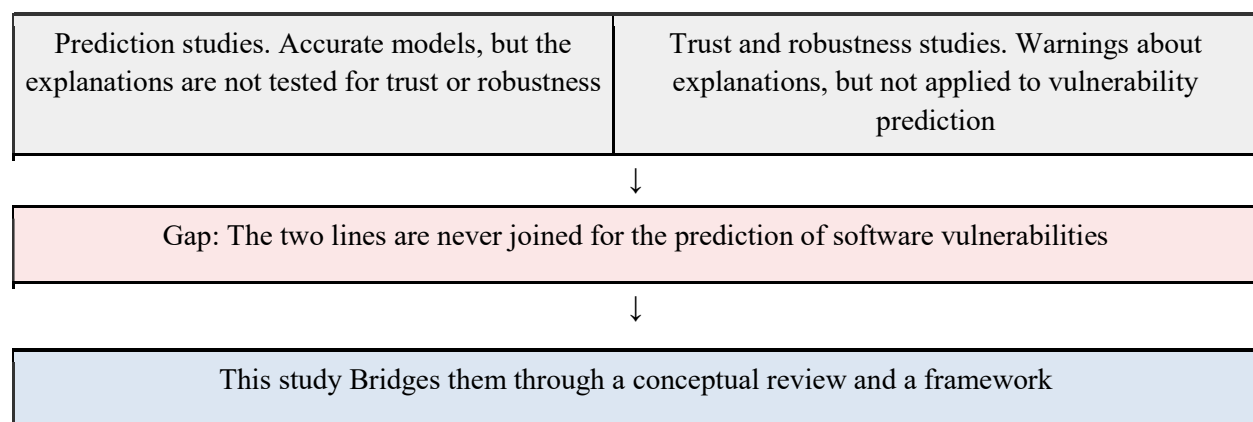


Figure 1. The research gap and how this study addresses it.

VI. Problem Statement

Although machine learning models have reached a very high accuracy in predicting software vulnerabilities, its decisions/outcomes often remain hard to understand, and this lack of transparency has limited how far security analysts can trust and act on them [9]. XAI has been presented as the solution, but the evidence is mixed. Some studies have shown that explanations improve understanding and support action [1], [6], while others have shown that explanations raise trust moderately [7], and still others have shown that the explanations can be attacked or deformed/twisted without the analyst noticing [2].

Because of all these, it is not properly clear whether explainable methods truly make the prediction of software vulnerabilities transparent and trustworthy, or whether they only create an appearance of clarity and transparency. This uncertainty is the problem the proposal addresses.

VII. Aim and Objectives

This study aims to examine how explainable artificial intelligence can make the prediction of software vulnerabilities more transparent and trustworthy. The aim is broken into four objectives.

Objective One: Reviewing the explainable methods currently used for predicting and assessing software vulnerabilities.

Objective Two: Analysing how these explanations support transparency and trust for the security analysts who largely rely on them.

Objective Three: Examining the limitations and risks of these explanations, including how robust they are against distortion or deformation.

Objective Four: Proposing a conceptual framework that links vulnerability prediction, explanation quality, transparency, and trust, in a form that a later study could test on a real vulnerability dataset.

Figure 2 shows how the single aim breaks into the four objectives.

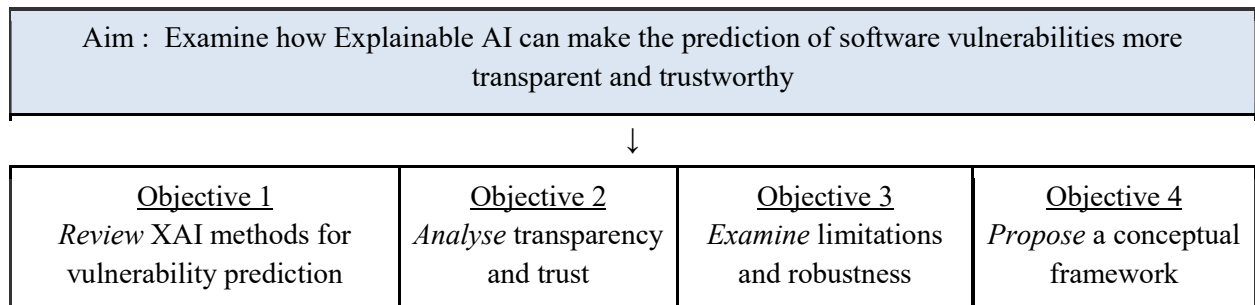


Figure 2. The aim is broken into four objectives.

VIII. Research Questions or Hypotheses

Because this study is a conceptual review rather than a test of variables, it is guided by research questions instead of hypotheses. The main research question comes directly from the title.

The main research question is “How can explainable artificial intelligence make the prediction of cybersecurity vulnerabilities in software systems more transparent and trustworthy?”

To answer this main question, the study addresses four supporting research questions.

RQ1. How can explainable artificial intelligence make the prediction of software vulnerabilities more transparent for security analysts?

RQ2. How does the quality of an explanation affect how much an analyst trusts a vulnerability prediction?

RQ3. What limitations or risks reduce the reliability of these explanations?

RQ4. What conceptual framework can connect explainable prediction, transparency, and trust in this setting?

IX. Conceptual Framework

The conceptual framework of this study connects four main ideas drawn from the literature review. The first idea is vulnerability prediction. This is the output of a ML model that judges whether a software weakness exists or how severe it is. The second idea is explanation quality. This is a description of how clear, complete, and faithful the reason given for that prediction is. The third idea is transparency. This gives us the degree to which an analyst can see and follow reasons behind the prediction. The fourth idea is trust. Finally this is the confidence the analyst places in the prediction before acting on it.

The framework proposes the following relationships. A vulnerability prediction on its own gives no to little transparency, because the model acts as a black box. When an explanation is added, transparency rises, because the analyst can see why the prediction has been made [1], [6]. Higher transparency is expected to support higher trust, but only up to a point, since trust also depends on whether the explanation is reliable [7]. The framework therefore adds sturdiness as a condition. If the explanation can be attacked or crooked, then transparency becomes misleading and trust is no longer justified [2], [9]. In short, explanation quality and robustness together decide whether transparency leads to real trust or only to the appearance of it.

This relationship is shown as a simple model in Figure 3. Here vulnerability prediction leads to an explanation. Then the explanation is shaped by quality and robustness leading to transparency, and finally transparency leads to trust. To note that, robustness here is acting as a gate on the final step. The framework is described here in words and as a model, and later evidence-based research could turn it into a testable model using a real vulnerability dataset.

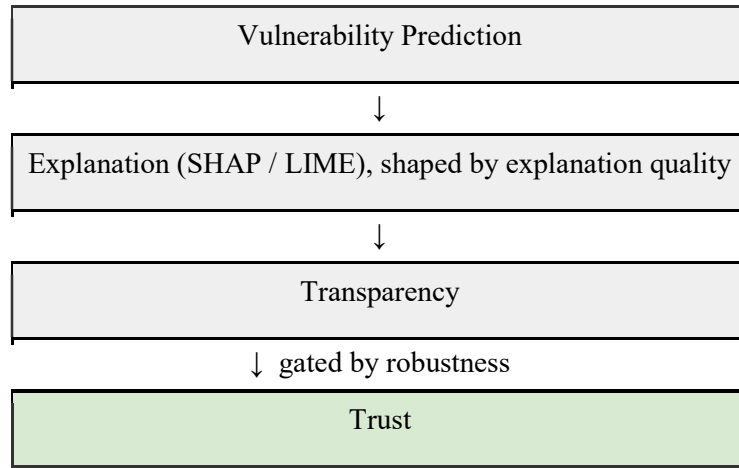


Figure 3. The proposed conceptual framework linking prediction, explanation, transparency, and trust.

X. Methodology

In this study I divide my methodology like below-

Research Design: This study uses a conceptual review. It does not collect new data or involve human participants. Instead, it studies a focused set of existing academic papers and articles to understand how explainable artificial intelligence supports the transparent and trustworthy prediction of software vulnerabilities. This design fits the review type of paper described in the proposal guide, where published articles are the main source and the analysis is carried out by theme.

Data Source: The data source is published academic literature in years of 2022 to 2026. The papers have been drawn from recognised academic databases and publishers, including IEEE, MDPI, and Elsevier, along with other peer reviewed venues, so that the sources are reliable and citable.

Selected Articles: The selected articles are eleven recent papers, published mainly between 2024 and 2026, that deal with explainable artificial intelligence, vulnerability prediction, threat detection, and trust in cybersecurity. They have been chosen because together they cover the prediction of vulnerabilities, the application of explanation methods, and the trust and robustness of those explanations, which are the themes this study needs.

Data Collection: It means the reading and recording of each paper. For every paper the study records its aim, its method, the explanation technique used, the security task addressed, its main findings, and its stated limitations.

Analysis: The analysis method is thematic analysis. The recorded information is grouped into themes, namely prediction, explanation method, transparency, trust, and robustness. By comparing the papers within and across these themes, the study identifies common findings, disagreements, and missing links, and uses them to build the conceptual framework. Through this analysis the study also shows, in a structured way, how explanation methods such as SHAP and LIME make a vulnerability prediction more transparent, which is a central concern of the proposal.

XI. Expected Contribution

This study is expected to make three contributions.

The theoretical contribution is a conceptual framework that links vulnerability prediction, explanation quality, transparency, and trust. And this joins two lines of research that have so far stayed apart [2], [7].

The practical contribution is strong guidance for security teams, analysts and tool designers, as the study clarifies when an explanation can be trusted and when it may mislead. This can help them choose and present explanations more carefully [1], [9].

The methodological contribution is the well-structured review and the comparison table (Table1), which organise a scattered body of work into clear themes and offer a base that a later in-depth study could build on using a real vulnerability dataset.

XII. Ethical Considerations

Because this study is based only on published academic sources, it does not involve human participants, personal data consent, or sensitive information. Therefore, the main ethical duties are about the responsible use of sources. All research sources are read fairly and reported accurately, and every idea taken from another work is credited through correct citation in order to avoid plagiarism. It has also been noted that the reviewed papers may carry their own bias, for example by reporting mainly positive results, and it takes this into account when drawing conclusions. Finally, the study supports the responsible use of artificial intelligence by promoting transparency and trust, which are ethical goals in security systems that affect people.

XIII. Timeline

The work has been divided into the stages shown below, and each stage is kept short so the proposal can be finished on time

Stage	Dates
Topic selection and planning	May 13 to May 15
Article selection and Reading	May 16 to May 22
Literature review	May 23 to May 31
Thematic analysis and framework development	June 1 to June 7
Writing the first draft	June 8 to June 14
Revision, formatting, and submission	June 15 to June 18

Table 2. Timeline of the Study.

XIV. Conclusion

This proposal has set out the path to research how explainable artificial intelligence can make the prediction of software vulnerabilities more transparent and trustworthy. It has reviewed eleven recent studies and has found a clear gap. That is first of all the prediction work. Secondly, studies of trust and robustness in explanations have stayed apart. To address these two gaps, the proposal has aimed to bring them together and to offer a conceptual framework that links prediction, explanation quality, transparency, and trust. The expected value is a much clearer understanding of the time when explanations can be trusted in vulnerability prediction, along with a base that a later empirical study could test on real data.

XV. References

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